

Answer Key — Biology Grade 11

Sub-Strand 1.3: Taxonomy II • Lesson 5: Kingdom Animalia — major phyla

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	C	B	B	B	B	C

USE AFTER: PREDICT PHASE — Predict: sorting the animal picture cards and recording rules on the board

Q1. In a class activity, groups sorted eight animal pictures. One group sorted by size, another by 'animals we eat', and a third by habitat. A scientist in Nairobi and one in Kisumu need to agree on how to group animals. What is the main problem with these sorting rules?

- A. They depend on opinion or situation, so different people get different groups**
- B. They cannot be written on manila paper
- C. They only work for animals that live on land
- D. They use too many animals at once

Correct answer: A. Size, food use and habitat vary with the person or place, so groupings differ. Scientists need shared, observable features that give the same result for everyone.

USE AFTER: OBSERVE PHASE — Observe: Expert A group (Annelida and Mollusca: earthworm, snail)

Q2. A learner says, 'A snail and an earthworm are both soft-bodied, so they must belong to the same phylum.' Which observation best shows the snail is NOT in the same phylum as the earthworm?

- A. The snail is smaller than the earthworm
- B. The snail is found in gardens, while the earthworm lives in soil
- C. The snail has a hard shell and an unsegmented body, while the earthworm has a body divided into many rings**
- D. The snail moves more slowly than the earthworm

Correct answer: C. Snails (Mollusca) have a soft unsegmented body with a shell, while earthworms (Annelida) have a segmented body with rings. Size, speed and habitat are not phylum features.

USE AFTER: OBSERVE PHASE — Observe: Expert B group (Arthropoda: butterfly, mosquito, grasshopper)

Q3. A mosquito has three body parts, six jointed legs and a hard outer covering. Which phylum does it belong to?

- A. Chordata
- B. Arthropoda**
- C. Annelida
- D. Mollusca

Correct answer: B. Jointed legs and a hard exoskeleton are features of Arthropoda. The mosquito has no backbone, no shell and no body rings.

USE AFTER: OBSERVE PHASE — Observe: Expert D and Expert E groups (Chordata: tilapia, lizard, chicken, goat)

Q4. A tilapia from Lake Victoria, a lizard, a chicken and a goat all look very different. Which feature do they ALL share that places them in the same phylum?

- A. Living in water
- B. A backbone**
- C. Scales covering the body
- D. Wings for flying

Correct answer: B. Chordata in this lesson are the animals with a backbone (fish, reptiles, birds, mammals). Their body coverings, habitats and movement differ.

USE AFTER: EXPLAIN PHASE — Explain: writing Claim-Evidence-Reasoning (CER) statements for chosen animals

Q5. A learner writes this claim: 'The butterfly belongs to Phylum Arthropoda.' Which piece of evidence best supports the claim?

- A. It is found in the school garden
- B. It has jointed legs and an exoskeleton**
- C. It is useful for pollinating flowers
- D. It is beautiful and colourful

Correct answer: B. Good evidence is an observable body feature that anyone can check. Jointed legs and an exoskeleton are Arthropoda features, while the other options are opinion, habitat or usefulness.

USE AFTER: EXPLAIN PHASE — Explain: comparing why the mosquito and goat are in different phyla

Q6. A mosquito feeds on blood and a goat feeds on grass, yet they are placed in different phyla. Which is the best reason?

- A. The mosquito can fly but the goat cannot
- B. The mosquito has jointed legs and an exoskeleton with no backbone, while the goat has a backbone**
- C. The mosquito is smaller than the goat
- D. The mosquito feeds on blood but the goat does not

Correct answer: B. Phyla are decided by body features such as a backbone, jointed legs and exoskeleton. The mosquito is an arthropod and the goat is a chordate. Size, feeding and flight are not used to decide phylum.

USE AFTER: MODEL BUILDING PHASE — Model: revising the class classification tree diagram with phyla and distinguishing features

Q7. A class is building a tree diagram of Kingdom Animalia. Which animal is placed correctly under its phylum?

- A. Butterfly under Chordata, because it can fly
- B. Lizard under Mollusca, because it is cold to touch
- C. Earthworm under Annelida, because its body is divided into rings and it has no legs**
- D. Snail under Arthropoda, because it has a hard shell

Correct answer: C. The earthworm's segmented body with no legs places it in Annelida. The snail is Mollusca, the lizard is Chordata and the butterfly is Arthropoda.